Use Cases

UC1: User Login/Logout/Create Account

Preconditions:
- The user has access to the store-facing system (web or app).
- The system is online and available.

Main Flow:
- The User opens the store-facing application.
- System presents authentication options [Display Authentication Options].
- User chooses to log in, log out, or create a new account.
  - If login is selected: User provides valid credentials [Login].
  - If logout is selected: User confirms logout [Logout].
  - If create account is selected: User provides required details according to the role. (Customer/Staff/Admin)[Create Account]
- System confirms the action [Confirm Authentication].

Subflows:
- [Display Authentication Options] System displays "Login," "Logout," and "Create Account."
- [Login] User enters username/email and password. System verifies credentials.
- [Logout] System prompts user to confirm logout. Upon confirmation, session ends.
- [Create Account] User enters name, email, password, and accepts terms. System validates inputs and stores the new user profile.
- [Confirm Authentication] System shows success message and updates session status accordingly.

Alternative Flows and Edge Cases:
- [AF1: Invalid Login] At [Login], if credentials are invalid, show: "Invalid username or password. Please try again." Allow retry or return to [Display Authentication Options].
- [AF2: Duplicate Account] At [Create Account], if the email is already registered, show: "An account with this email already exists." Redirect to [Login].
- [AF3: Weak Password] At [Create Account], if password does not meet strength requirements, show: "Password must include letters, numbers, and a special character." Prompt re-entry.
- [AF4: Session Timeout] After login, if the session remains idle beyond the timeout threshold, system automatically logs the user out and returns to [Display Authentication Options].
- [AF5: Logout Cancelled] At [Logout], if user cancels logout, system keeps the current session active and returns to the main dashboard.

UC2: Restaurant Profile Management

Preconditions:

- The restaurant user is logged in.
- The restaurant profile exists in the system.

Main Flow:

- The Restaurant User navigates to the profile management section.
- System displays the current restaurant profile [Display Profile].
- User selects “Edit Profile” [Edit Profile].
- User updates fields such as name, address, hours, cuisine type, or contact information [Update Information].
- User saves changes [Save Profile].
- System validates and stores the updates [Confirm Update].

Subflows:

- [Display Profile] System retrieves and shows current restaurant profile details.
- [Edit Profile] User chooses to modify profile fields.
- [Update Information] User enters new or updated information in one or more fields.
- [Save Profile] User confirms changes by selecting “Save.”
- [Confirm Update] System validates input, saves changes, updates the profile, and shows a success message.

Alternative Flows and Edge Cases:

- [AF1: Invalid Input] At [Update Information], if required fields are missing or invalid (e.g., incorrect phone format), show: “Please correct highlighted fields.” Return to [Edit Profile].
- [AF2: Duplicate Restaurant Name] At [Save Profile], if the name conflicts with another registered restaurant, show: “This restaurant name is already taken.” Return to [Edit Profile].
- [AF3: Save Cancelled] At [Save Profile], if the user cancels instead of confirming, no changes are made and system returns to [Display Profile].
- [AF4: System Error] At [Confirm Update], if the system cannot save due to technical error, show: “Unable to save changes at this time. Please try again later.” Return to [Display Profile].
- [AF5: No Permission] If a staff user without admin privileges attempts to edit the profile, show: “You do not have permission to edit this information.” Return to dashboard.

<u>**UC3: Inventory Management**</u>

Preconditions:

- The restaurant user is logged in with appropriate permissions.
- The restaurant profile is active in the system.

Main Flow:

- The Restaurant User navigates to the inventory management section.
- System displays the current list of inventory items [Display Inventory].
- User selects an action: add, update, or remove an item.

    - If add is selected: User provides details of the new item [Add Item].
    - If update is selected: User modifies existing item details [Update Item].
    - If remove is selected: User confirms deletion [Remove Item]
- User saves the changes [Save Inventory].
- System validates and applies updates [Confirm Inventory Update].

Subflows:

- [Display Inventory] System retrieves and shows the current inventory with quantities and status.
- [Add Item] User enters item name, quantity, unit type, and optional supplier info.
- [Update Item] User edits existing item details (e.g., quantity, price, availability).
- [Remove Item] System asks for confirmation; upon confirmation, removes the item.
- [Save Inventory] User confirms action to commit changes.
- [Confirm Inventory Update] System validates input, updates records, and shows a success message.

Alternative Flows and Edge Cases:

- [AF1: Invalid Input] At [Add Item] or [Update Item], if a field is invalid (e.g., negative quantity), show: “Invalid input. Please check fields.” Return to editing.
- [AF2: Duplicate Item] At [Add Item], if the item already exists, show: “This item is already in inventory. Update instead?” Redirect to [Update Item].
- [AF3: Remove Cancelled] At [Remove Item], if the user cancels, item remains in inventory and system returns to [Display Inventory].
- [AF4: Low Stock Alert] At [Update Item], if quantity falls below threshold, system flags item as “Low Stock” and notifies user.
- [AF5: System Error] At [Confirm Inventory Update], if the system cannot save due to technical issue, show: “Unable to save changes. Please try again later.” Return to [Display Inventory].
- [AF6: No Permission] If a non-manager attempts to manage inventory, system shows: “You do not have permission to edit inventory.” Redirect to dashboard.

<u>UC4: Discount/Sale Promotion</u>

Preconditions:

- The restaurant user is logged in with manager/admin permissions.
- The restaurant profile is active in the system.

Main Flow:

- The Restaurant User navigates to the promotions section.
- System displays current active and scheduled promotions [Display Promotions].
- User selects “Create Promotion” or chooses an existing promotion to edit/remove.
    - If create: User enters details such as discount type, percentage, eligible items, start/end dates [Create Promotion].
    - If edit: User modifies existing promotion details [Edit Promotion].

    - If remove: User confirms deletion [Remove Promotion].
- User saves changes [Save Promotion].
- System validates and applies updates [Confirm Promotion].

Subflows:
- [Display Promotions] System shows all current and upcoming promotions.
- [Create Promotion] User defines promotion type (e.g., “20% off all desserts”), timeframe, and conditions.
- [Edit Promotion] User changes details of an existing promotion.
- [Remove Promotion] System asks for confirmation before deletion.
- [Save Promotion] User confirms action to commit changes.
- [Confirm Promotion] System validates inputs, updates database, and displays confirmation.

Alternative Flows and Edge Cases:
- [AF1: Invalid Dates] At [Create Promotion], if start date is after end date, show: “Invalid promotion dates.” Return to editing.
- [AF2: Overlapping Promotions] At [Save Promotion], if a conflict exists with another active promotion, system warns: “This overlaps with an existing promotion. Continue?” Allow user to confirm or cancel.
- [AF3: Remove Cancelled] At [Remove Promotion], if user cancels, system keeps the promotion active and returns to [Display Promotions].
- [AF4: System Error] At [Confirm Promotion], if system cannot save changes, show: “Unable to save promotion. Please try again later.” Return to [Display Promotions].
- [AF5: No Permission] If a staff user without manager rights attempts access, show: “You do not have permission to manage promotions.” Redirect to dashboard.

UC5: Order Management

Preconditions:
- The restaurant user is logged in.
- Orders exist in the system (pending, in-progress, or completed).

Main Flow:
- The Restaurant User navigates to the order management section.
- System displays current orders with statuses [Display Orders].
- User selects an order to review [Select Order].
- User updates the order status (e.g., ACCEPTED, PREPARING, READY, OUT FOR DELIVERY, COMPLETED) [Update Order Status].
- User may add internal notes or adjust estimated preparation time [Add Notes/ETA].
- System saves updates and notifies customer/driver as applicable [Confirm Order Update].

Subflows:
- [Display Orders] System shows all active and historical orders with filters.
- [Select Order] User chooses an individual order to manage.

- [Update Order Status] User changes the current status of the order.
- [Add Notes/ETA] User enters optional notes or updated preparation time.
- [Confirm Order Update] System validates changes, applies updates, and sends real-time notifications.

Alternative Flows and Edge Cases:

- [AF1: Order Already Cancelled] At [Update Order Status], if the order was cancelled by the customer, show: “This order has been cancelled.” Return to [Display Orders].
- [AF2: Invalid Status Transition] If user attempts to skip steps (e.g., directly from ACCEPTED to COMPLETED), system prevents update and shows: “Invalid status change.”
- [AF3: System Error] At [Confirm Order Update], if system cannot save, show: “Unable to update order at this time.” Return to [Select Order].
- [AF4: Driver Unavailable] At [Update Order Status → OUT FOR DELIVERY], if no driver is available, show: “No driver currently available. Please try again shortly.” Return to [Select Order].
- [AF5: Unauthorized User] If a staff member without order permissions attempts an update, show: “You do not have permission to manage orders.” Redirect to dashboard.

UC6: Real-Time Contact with Customer or Driver

Preconditions:

- The restaurant user is logged in.
- At least one active order exists with a linked customer or driver.

Main Flow:

- The Restaurant User opens the communication panel for an active order.
- System displays contact options (chat, call, in-app message) [Display Contact Options].
- User selects the desired method of contact [Select Contact Method].
- User initiates the communication [Initiate Contact].
- System establishes a real-time channel and logs interaction details [Confirm Contact].

Subflows:

- [Display Contact Options] System shows available methods based on order and privacy settings.
- [Select Contact Method] User chooses chat, call, or in-app message.
- [Initiate Contact] User types message or places call.
- [Confirm Contact] System confirms contact channel is open and records the interaction in the order history.

Alternative Flows and Edge Cases:

- [AF1: Customer Unresponsive] At [Initiate Contact], if customer does not respond within a set time, system notifies: “Customer is not responding.” Option to retry or cancel.
- [AF2: Driver Unavailable] At [Initiate Contact], if driver cannot be reached, system shows: “Driver currently unavailable.” Option to retry or assign backup contact method.

- [AF3: Communication Restricted] If user attempts to contact outside permitted hours, system shows: “Contact not available at this time.”
- [AF4: Technical Failure] If chat/call fails to connect, system shows: “Unable to connect. Please try again.” Return to [Display Contact Options].
- [AF5: Unauthorized Access] If a staff member without permission attempts to initiate contact, system blocks action with: “You do not have permission to contact customers/drivers.”

UC7: Order Queue/Ticketing System

Preconditions:
- The restaurant user is logged in.
- At least one new order exists in the system.

Main Flow:
- The Restaurant User navigates to the order queue section.
- System displays all active orders in a ticket-style view [Display Queue].
- Orders appear sorted by order time or priority.
- User selects a ticket to view details [Select Ticket].
- User updates ticket status (e.g., In Preparation, Ready) [Update Ticket].
- System updates the queue and notifies relevant parties [Confirm Queue Update].

Subflows:
- [Display Queue] System shows active orders as individual tickets, including order number, items, and status.
- [Select Ticket] User taps or clicks a ticket to expand details.
- [Update Ticket] User marks the ticket with the appropriate status.
- [Confirm Queue Update] System validates status change, updates display, and notifies customer/driver.

Alternative Flows and Edge Cases:
- [AF1: No Active Orders] At [Display Queue], if there are no active orders, system shows: “No orders in queue.”
- [AF2: Ticket Already Completed] At [Update Ticket], if the ticket is marked completed, show: “This ticket is closed.” Return to [Display Queue].
- [AF3: Invalid Update] If user attempts to revert a ticket back to NEW after processing, system blocks and shows: “Invalid action.”
- [AF4: System Error] At [Confirm Queue Update], if update fails, show: “Unable to update ticket. Please retry.” Return to [Display Queue].

UC8: Customer Service / Feedback

Preconditions:

- The restaurant user is logged in.
- At least one order exists with customer feedback enabled.

Main Flow:

- The Restaurant User navigates to the feedback section.
- System displays recent customer reviews and ratings [Display Feedback].
- User selects a feedback entry to view details [Select Feedback].
- User may respond to the feedback [Respond to Feedback].
- System records and publishes the response (if public) [Confirm Response].

Subflows:

- [Display Feedback] System shows customer comments, star ratings, and order references.
- [Select Feedback] User views a specific feedback entry.
- [Respond to Feedback] User writes a reply or takes internal action (e.g., flag for review).
- [Confirm Response] System saves reply, links it to the feedback, and updates record.

Alternative Flows and Edge Cases:

- [AF1: No Feedback Available] At [Display Feedback], if no feedback exists, system shows: "No customer feedback yet."
- [AF2: Inappropriate Response] At [Respond to Feedback], if response violates platform policy, system blocks and shows: "Your response cannot be posted. Please revise."
- [AF3: Feedback Already Closed] At [Respond to Feedback], if feedback thread is closed, system shows: "This feedback cannot be updated."
- [AF4: System Error] At [Confirm Response], if saving fails, system shows: "Unable to post response. Try again later."

<u>UC9: Export Sales/Order Reports</u>

Preconditions:

- The restaurant user is logged in with manager/admin permissions.
- Sales and order history exists in the system.

Main Flow:

- The Restaurant User navigates to the reporting section.
- System displays reporting options (date range, report type) [Display Report Options].
- User selects desired filters and format (CSV, PDF, XLS) [Select Report Parameters].
- User requests export [Export Report].
- System generates report file [Generate Report].
- User downloads or views the report [Confirm Export].

Subflows:

- [Display Report Options] System shows available report types (sales summary, item sales, order history).
- [Select Report Parameters] User selects filters such as timeframe, order type, or payment method.
- [Export Report] User submits export request.

- [Generate Report] System queries data, formats report, and prepares file.
- [Confirm Export] System provides download link or display.

Alternative Flows and Edge Cases:

- [AF1: No Data Found] At [Generate Report], if no orders match filters, system shows: “No data available for selected criteria.”
- [AF2: Invalid Parameters] At [Select Report Parameters], if input is invalid (e.g., end date before start date), system shows error and returns to selection.
- [AF3: Large Report Delay] At [Generate Report], if data is too large, system shows: “Report is being generated. You will be notified when ready.”
- [AF4: System Error] At [Confirm Export], if file cannot be generated, show: “Export failed. Please try again later.”

UC10: Customer Cancels Order

Preconditions:

- The restaurant user is logged in.
- At least one active customer order exists.

Main Flow:

- The Customer cancels the order from their app.
- System updates the order status to CANCELLED [Cancel Order].
- Restaurant User receives cancellation notification [Notify Restaurant].
- Restaurant User reviews cancelled order details [Review Cancellation].
- System adjusts inventory or refunds payment as applicable [Process Cancellation].

Subflows:

- [Cancel Order] Customer cancels order in their app.
- [Notify Restaurant] System notifies restaurant with order ID and cancellation reason (if provided).
- [Review Cancellation] User views cancelled order in the order list.
- [Process Cancellation] System updates inventory, processes refund (if prepaid), and closes order.

Alternative Flows and Edge Cases:

- [AF1: Cancellation Too Late] At [Cancel Order], if order is already being delivered, system blocks cancellation and notifies customer: “This order cannot be cancelled at this stage.”
- [AF2: Refund Failed] At [Process Cancellation], if refund transaction fails, system flags issue and notifies restaurant user to resolve manually.
- [AF3: Order Already Completed] At [Cancel Order], if order status is COMPLETED, system shows: “This order cannot be cancelled.”
- [AF4: Notification Failure] At [Notify Restaurant], if restaurant cannot be notified (e.g., system offline), system logs event and retries later.

UC11: Automated Inventory Reordering

Preconditions:
- The restaurant user is logged in with manager permissions.
- Inventory thresholds are configured in the system.

Main Flow:
- System monitors stock levels in real-time.
- When an item drops below the threshold, system creates a suggested supplier order [Generate Reorder Suggestion].
- Restaurant User reviews the reorder suggestion [Review Suggestion].
- User confirms or modifies the order [Confirm Reorder].
- System places the order with supplier or records it for manual fulfillment [Process Reorder].

Subflows:
- [Generate Reorder Suggestion] System calculates needed quantities based on sales trends.
- [Review Suggestion] User views auto-generated supplier order.
- [Confirm Reorder] User approves, edits, or cancels the suggestion.
- [Process Reorder] System sends purchase order to supplier (if integrated) or saves record.

Alternative Flows and Edge Cases:
- [AF1: No Supplier Linked] If no supplier exists for the item, system shows: "No supplier available. Please contact supplier manually."
- [AF2: Suggestion Rejected] If user rejects the suggestion, order is not placed, system logs event.
- [AF3: Supplier Error] At [Process Reorder], if supplier integration fails, system notifies user and saves order for manual action.

UC12: Time-Bound Promotions (Happy Hour)

Preconditions:
- The restaurant user is logged in.
- At least one promotion has been scheduled in the system.

Main Flow:
- Restaurant User navigates to the promotions section.
- User configures a time-bound discount [Set Time Promotion].
- System activates promotion automatically at scheduled time [Activate Promotion].
- During active period, discounted items appear in the menu [Apply Promotion].
- At end time, promotion deactivates [End Promotion].

Subflows:
- [Set Time Promotion] User selects items, discount %, start and end times.
- [Activate Promotion] System updates availability and pricing at scheduled time.
- [Apply Promotion] Orders placed during active period use discounted prices.

- [End Promotion] System restores normal pricing automatically.

Alternative Flows and Edge Cases:

- [AF1: Overlapping Promotions] If two promotions overlap, system warns and requires resolution.
- [AF2: Early Cancellation] User manually ends promotion before end time, system restores pricing immediately.
- [AF3: Invalid Schedule] If end time is before start time, system blocks and shows error.

UC13: Scheduled Export of Reports

Preconditions:

- The restaurant user is logged in with reporting permissions.
- Reporting schedule has been configured.

Main Flow:

- Restaurant User navigates to reporting settings.
- User schedules automatic report exports [Schedule Export].
- System generates report at scheduled intervals [Generate Report].
- System delivers report via email or in-app notification [Deliver Report].
- Restaurant User downloads or views report [Access Report].

Subflows:

- [Schedule Export] User sets frequency (daily, weekly, monthly), report type, and format.
- [Generate Report] System runs query and prepares report file.
- [Deliver Report] System sends file to designated channel.
- [Access Report] User opens report link or downloads file.

Alternative Flows and Edge Cases:

- [AF1: Invalid Schedule] If frequency is invalid (e.g., multiple overlapping reports), system shows error.
- [AF2: Delivery Failure] If email or push fails, system retries and logs issue.
- [AF3: No Data Found] If report has no data, system sends file with “No data available” notice.

UC14: Negative Feedback Compensation

Preconditions:

- The restaurant user is logged in.
- At least one customer feedback entry exists with a negative rating.

Main Flow:

- Restaurant User reviews feedback [Review Feedback].
- System suggests possible compensations (e.g., discount coupon) [Suggest Compensation].
- User selects and confirms compensation [Confirm Compensation].
- System delivers compensation to customer [Deliver Compensation].

- System logs resolution [Record Resolution].

Subflows:

- [Review Feedback] User opens negative feedback entry.
- [Suggest Compensation] System recommends options based on severity (e.g., free dessert, % off next order).
- [Confirm Compensation] User chooses offer and approves.
- [Deliver Compensation] System issues coupon or credit to customer's account.
- [Record Resolution] System stores resolution details for reporting.

Alternative Flows and Edge Cases:

- [AF1: No Action Taken] If user decides not to compensate, system records "No action" resolution.
- [AF2: Compensation Failed] If coupon/credit cannot be issued, system shows: "Failed to deliver compensation. Retry later."
- [AF3: Duplicate Compensation] If compensation already issued, system blocks duplicate issue.

Reflection

As our team reviewed the totality of the use cases we had made, we determined that an entire Food Delivery Application was too ambitious an endeavor for a timeframe of eight weeks. As such, we decided to focus on the business aspect of such an application. This involves scrapping any features that were focused on the Customer or Delivery Drivers, such as Order Placement and Payment, and Order Tracking. For now, the primary features for the Minimum Viable Product will revolve around the following points:

1. Restaurant Profile Management
2. User Account Management
3. Inventory Management
4. Discount Promotion
5. Order Management and Queue Tracking
6. Customer Service / Customer Contact
7. Report Generation and Export

These features will provide a Product that can, at a minimum, be used by Restaurants to manage and process one-to-many Orders placed, as well as being able to manage and view any necessary information regarding the restaurant, including the public profile and financial reports.

While beneficial to the Restaurant stakeholders, this Minimum Viable Product will disappoint the customer and delivery driver stakeholders as they will not be the primary focus and are not included in the Minimum Viable Product. The primary focus under the Customer stakeholder was Customer Access, and Customer's being able to manage their Order. Similarly, the Delivery Driver stakeholder is not included in this Minimum Viable Product and will not have access to the platform and will not be able to track and manage Order Deliveries. Additionally, the primary stakeholders we are focusing on, the Restaurant Staff, will have justifiable concerns over potential loss of functionality previously discussed during the initial planning stages of the platform.

The issue with downsizing the platform requirements was the loss of granularity which satisfied stakeholders, and was beneficial for testing and traceability. However, by downsizing the platform, key requirements have been identified, while other requirements previously considered can be satisfied in future iterations or development. To alleviate the concerns of all stakeholders, the Minimum Viable Product must be developed as a modular platform. By allowing modularity, more features can be added, or the platform can be integrated into a larger project. Modularity for this platform will be implemented through the use of an API, allowing for both future applications or features to more easily interact with the product, while retaining core functionality. While this does not directly address concerns regarding features, it opens a pathway for the stakeholders to be later included through iterative development and improvement.

LLM Receipts

https://chatgpt.com/share/68c37555-44f0-8001-9509-7abcfa3b1a7c